

Catherine (Catie) Wiener

PHD STUDENT · EPIDEMIOLOGY

Gillings School of Public Health, University of North Carolina at Chapel Hill

cwiener@live.unc.com | ktwiener.github.io

Education

University of North Carolina at Chapel Hill

Chapel Hill, NC

PHD EPIDEMIOLOGY

August 2022 - present

- Advisor: Dr. Stephen R. Cole

MS BIostatISTICS

August 2017 - May 2019

- Advisor: Dr. Gary G. Koch
- Thesis: A robust model to detect disease modification for a delayed-start design

BS MATHEMATICS & ENGLISH

August 2010 - August 2014

- Minor in Mathematical Decision Sciences

Experience

- 2024- **Graduate Research Assistant**, MACS/WIHS Combined Cohort Study, University of North Carolina at Chapel Hill
- 2022-2024 **Research Fellow**, Center for Pharmacoepidemiology, UNC Gillings School of Global Public Health
- 2019-2022 **Statistician**, Target RWE
- 2017-2019 **Graduate Research Assistant**, Department of Biostatistics, University of North Carolina at Chapel Hill
- 2014-2017 **Statistical Programmer**, Quintiles/Quintiles IMS
- 2010-2014 **Undergraduate Research Assistant**, Department of Biostatistics, University of North Carolina at Chapel Hill

Teaching

University of North Carolina at Chapel Hill

Chapel Hill, NC

GRADUATE TEACHING ASSISTANT

2024-2026

- EPID 717: Advanced Epidemiologic Methods
- EPID 700: Foundations of Scientific Reasoning in Epidemiology (Instructor for weekly R programming lectures)
- EPID 701: R Programming for Epidemiologists

40th Annual Meeting of the International Society for Pharmacoepidemiology

Berlin, Germany

WORKSHOP SPEAKER

2024

- Using generative AI to more efficiently conduct systematic reviews, solve coding problems, and communicate research findings: a student-led workshop

2025 Society for Epidemiologic Research Conference

Boston, MA

WORKSHOP SPEAKER

2025

- Modern use of propensity scores: weighting, estimands, and moving beyond the bootstrap for variance estimation.

Awards and Honors

- 2026 **Gillings School Graduate Teaching Assistant Award**, Gillings School of Public Health, UNC-CH
- 2026 **H. Michael and Barbara Arrighi Endowed Scholarship**, Epidemiology, Gillings School of Public Health, UNC-CH
- 2025 **Anne Cynthia Price Fund for Student Travel**, Gillings School of Public Health, UNC-CH
- 2024 **Scholarship for Annual Meeting**, International Society for Pharmacoepidemiology
- 2022 **Spotlight Poster Award**, International Society for Pharmacoepidemiology
- 2022 **Research Fellowship**, Center for Pharmacoepidemiology, Gillings School of Public Health, UNC-CH

Professional Development

- 2024-2025 **President**, UNC Student Chapter of the International Society for Pharmacoepidemiology
2024 **Volunteer**, Society for Epidemiological Research Annual Conference, Austin, Texas
2023- **Member**, Society for Epidemiological Research
2020- **Member**, International Society for Pharmacoepidemiology

Publications

* co-first authors

PREPRINTS

1. **Wiener, C.***, Latour, C.D.*, Hurwitz, K., Li, X., Lesko, C. R., Breskin, A. & Brookhart, M. A. Comparing Causal Estimands from Sequential Nested versus Single Point Target Trials: A Simulation Study. arXiv: 2601.20725 [stat.AP] (2026).

PUBLISHED

2. **Wiener, C.**, Edmonds, A., Sha, B., Stosor, V., Barjaktarevic, I., McCormack, M., Dionne, J., *et al.* Associations between Serological Evidence of SARS-CoV-2 Infection and Longitudinal Pulmonary Outcomes among People with HIV: Analysis of the MACSWIHS Combined Cohort Study. *Respiratory Research*. **In press**.
3. **Wiener, C.**, Zivich, P. N., Kurth, T., Jonsson-Funk, M., Breskin, A., Berger, K. & Cole, S. R. Causal Identification Conditions for the Effect of Treatment in the Treated: Illustration Using the Northwest Germany Stroke Registry. *Epidemiology* **37**, 57–66. ISSN: 1531-5487 (2026).
4. Latour, C. D., Delgado, M., Su, I. H., **Wiener, C.**, Acheampong, C. O., Poole, C., Edwards, J. K., *et al.* Use of Sensitivity Analyses to Assess Uncontrolled Confounding from Unmeasured Variables in Observational, Active Comparator Pharmacoepidemiologic Studies: A Systematic Review. *American Journal of Epidemiology*, kwae234. ISSN: 0002-9262 (2024).
5. Breskin, A*, **Wiener, C.***, Adimora, A. A., Brown, R. S. J., Landis, C., Reddy, K. R., Verna, E. C., *et al.* Effectiveness of Remdesivir Treatment Protocols Among Patients Hospitalized with COVID-19: A Target Trial Emulation. *Epidemiology* **34**, 365. ISSN: 1044-3983 (2023).
6. Levintow, S. N., Reading, S. R., Noshad, S., Mayer, S. E., **Wiener, C.**, Eledath, B., Exter, J., *et al.* Lipid Testing Trends Before and After Hospitalization for Myocardial Infarction Among Adults in the United States, 2008–2019. *Clinical Epidemiology* **14**, 737–748. ISSN: null (2022).
7. Cholera, R., Anderson, D., Raman, S. R., Hammill, B. G., DiPrete, B., Breskin, A., **Wiener, C.**, *et al.* Medicaid Coverage Disruptions Among Children Enrolled in North Carolina Medicaid From 2016 to 2018. *JAMA Health Forum* **2**, e214283. ISSN: 2689-0186 (2021).
8. Wang, D., Robieson, W., Zhao, J., **Wiener, C.** & Koch, G. Statistical Considerations in a Delayed-Start Design to Demonstrate Disease Modification Effect in Neurodegenerative Disorders. *Pharmaceutical Statistics* **18**, 407–419. ISSN: 1539-1612 (2019).

Presentations

* presenting author

1. **Wiener C.*** Changes in pulmonary function after SARS-CoV-2 infection among MWCCS participants with and without HIV. Invited Poster Presentation at the MWCCS Executive Committee Meeting Early Stage Investigator Meeting, Washington D.C., 4 December 2024.
2. Latour CD, **Wiener C.***, Breskin A, Hurwitz K, Brookhart MA. Comparing causal estimands from a sequential nested trial emulation with a treatment decision design to a conventional single point randomized trial: A simulation study. Oral presentation at the 40th Annual Meeting of the International Society for Pharmacoepidemiology (ISPE), Berlin, Germany, 24–28 August 2024.
3. **Wiener C.***, Cole SR, Kurth T, Berger K. Identifying the average treatment effect in the treated: Illustration using the Northwest Germany Stroke Registry. Poster presentation at the 40th Annual Meeting of the International Society for Pharmacoepidemiology (ISPE), Berlin, Germany, 24–28 August 2024.
4. **Wiener C.***, Cole SR. Identification conditions for the effect of treatment in the treated. Oral presentation at the 57th Annual Society for Epidemiologic Research (SER) Meeting. June 18–21, 2024. Austin, TX.
5. **Wiener C.***, Brookhart MA. An analytic method to weight longitudinal treatment patterns for visualization with Sankey diagrams. Oral presentation at the 39th International Conference on Pharmacoepidemiology and Therapeutic Risk Management (ICPE), Halifax, Canada, 25–27 August 2023. *Pharmacoepidemiol Drug Saf*, 32: 3-612.

6. **Wiener C**, Breskin A, Brookhart MA. Discharged where? The importance of understanding discharge dispositions when estimating in-hospital mortality. Poster presentation at the 38th International Conference on Pharmacoepidemiology & Therapeutic Risk Management, August 24-28, 2022
7. Breskin A, **Wiener C**, Brookhart MA. The effect of remdesivir for the treatment of COVID-19 on mortality among hospitalized patients. Oral presentation at the 38th International Conference on Pharmacoepidemiology & Therapeutic Risk Management, August 24-28, 2022
8. **Wiener C**, Breskin A, Rathnayaka N, Cook A, Brookhart MA. In-hospital treatment patterns among patients hospitalized with COVID-19 in the United States. Poster presentation at the 38th International Conference on Pharmacoepidemiology & Therapeutic Risk Management, August 24-28, 2022 *First place Poster Spotlight Session - Drug Utilization Research*
9. Rathnayaka N, Breskin A, Cook A, **Wiener C**, Brookhart MA. Risk factors for mortality and intensive care requirements among pediatric patients hospitalized with COVID-19 in the United States. *Pharmacoepidemiology and Drug Safety* 2022;58-9.
10. Levintow SN, Reading SR, Noshad S, Mayer S, **Wiener C***, Eledath B, Palagashvili T, Brookhart MA. Lipid testing trends before and after hospitalization for myocardial infarction among adults in the U.S., 2008-2019. Oral presentation at the ESC Congress, Virtual, August 27-30, 2021
11. **Wiener C***, Levintow SN, Orroth KK, Gill K, Mullard A, Brookhart MA, McGrath LJ. Real world treatment patterns following the initiation of proprotein convertase subtilisin/kexin type 9 antibody inhibitors among adults in the United States, 2015-2019. Poster presentation at the 36th International Conference on Pharmacoepidemiology & Therapeutic Risk Management, Virtual, September 16-17, 2020
12. Kim M, Park A, McGrath L, **Wiener C**, Balasubramania A, McDermott M, Deignan C, Brookhart MA. Trends in Osteoporosis Treatment Uptake and Persistence Among Postmenopausal Women in the U.S., 2010-2015. *Journal of the Endocrine Society* 2020;4(Supplement 1)SUN-389.
13. Koch Nicholas B., Kearney EK, **Wiener C**, Koch GG, Lorbacher LD, Viprakasit D. A comparison of patient desired outcomes between pain and non-pain-focused patients following nephrolithiasis surgery. *Journal of Urology* 2019;201(Supplement 4):e263-e263.
14. Narang G, Stephenson E, **Wiener C**, Koch G, Viprakasit D. Effectiveness of preventative counseling in recurrent stone formers. *Journal of Urology* 2019;201(Supplement 4):e174-e174.

Contributed Reviews

2025: American Journal of Epidemiology (2), BMC Infectious Disease (1), Epidemiology (1), Journal of Acquired Immune Deficiency Syndromes (1)

2026: American Journal of Epidemiology (1), Annals of Epidemiology (1), Journal of Acquired Immune Deficiency Syndromes (1)